Project Name: Document Scanning and Digitization

Index No: 200501P

In this project, my primary responsibility was to develop an Expo mobile application for capturing, editing, and sending photos to the backend. Additionally, I was tasked with creating a dashboard for rendering insights derived from the marked documents. As a part of the data science component, I focused on layout detection and exclusively rebuilding text.

Challenges Faced and Overcoming Them:

Data Collection: Gathered data from various sources and synthesized documents due to insufficient training data.

Text Rebuilding and Layout Detection: For text rebuilding, I utilized Pytesseract. For layout detection, after experimenting with various other tools like Layout Parser and Monk AI, which didn't yield the required accuracy. To improve accuracy, my partner and I trained a YOLO model, employing the Labelling tool for data annotation and IOU for accuracy measurement.

React Native Development: As it was my first time developing a React Native application, I encountered various issues while downloading dependencies. To overcome this, I switched to using 'yarn' instead of 'npm' for more reliable dependency management.

Styling Challenges: I faced difficulties in correctly placing components and styling the pages, which was particularly challenging. To tackle this, I dedicated time to understanding the intricacies of the styling framework, eventually achieving the desired layout.

Compatibility Issues: Some of the libraries and vision tools initially didn't support React Native and Expo. To resolve this, I explored alternative options and integrated suitable libraries that were compatible with the project requirements.

Backend-Frontend Integration: Connecting the front end to the backend posed a significant challenge, especially because both were on different PCs. Although initially lacking sufficient knowledge, we consulted various documents and resources, ultimately resolving the connectivity issues.

Limited Library Support: I encountered constraints with certain libraries due to the limitations of React Native Expo. To address this, I focused on exploring and incorporating compatible libraries that aligned with the project's requirements.

Despite the challenges, I effectively managed to contribute to the project by employing a combination of research, perseverance, and effective problem-solving strategies